

Mete Aksoy

Software Engineer, systems, backend & low-latency

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Education

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- Delft University of Technology**, MSc in Computer Science and Engineering Delft, NL
- **Coursework:** Distributed Systems, Cybersecurity
- Delft University of Technology**, BSc in Computer Science and Engineering Delft, NL
- **Coursework:** Object-Oriented & Functional Programming, Algorithms & Data Structures, Computer Networks, Computer Security, Concurrency & Operating Systems

Work Experience

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- Software Engineering Intern**, Dutch Ministry of Defense, KIXS IT Innovation Unit Amersfoort, NL
- Developed AR-HUD systems on the **Magic Leap 2** platform; implemented spatial anchoring for data overlays in dynamic, low-light conditions using **Unity**, integrated with **HLA-based** simulation software via **Pitch pRTI**.
- Software Engineering Intern**, PYM Automotive Istanbul, TR
- Supported development and automation of software tools for automotive remanufacturing machinery, improving system efficiency and data-tracking throughput by 8%.
 - Partnered with mechanical engineers to enhance machinery diagnostics and design workflows using **Python** and CAD-based tooling.

Projects

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- Specification-Based Fault Localization for Multi-Agent Systems** TU Delft, NL
- Designed and implemented a 6-stage LLM pipeline to automatically localize failures in multi-agent traces, combining specification extraction, dynamic constraint generation, and LLM-as-a-judge attribution.
 - Built custom trace normalizers for 6 heterogeneous log formats and a structured validation log used as evidence for final attribution.
- LOB-Engine** (Limit Order Book)
- Built a high-performance limit order book in **C++**, optimized for $O(1)$ order cancellations and $O(\log N)$ insertions across multiple order types.
- SPSC Ring Buffer**
- Built a lock-free, wait-free Single-Producer/Single-Consumer ring buffer in **C++20**, targeting low-latency **HFT** workloads on **x86-64**, eliminating cross-core cache-coherence traffic via a cached-index optimization.
 - Prevented false sharing by isolating producer and consumer state onto separate **64-byte cache lines**.
- Nexus** (API Gateway)
- Built a production-grade API gateway in **Go**, exposing a unified **GraphQL** and REST API over GitHub and **IBKR Client Portal**, with Redis caching and singleflight stampede prevention.
 - Implemented JWT auth with refresh-token rotation, sliding-window rate limiting, and a **circuit breaker** to protect upstream services under load.
- Housing Monitor Bot**
- Built an automated housing-listing monitor in **Python**, detecting new entries via API polling and sending real-time email alerts via **AWS EC2**; integrated **Selenium** for automated applications after `robots.txt` compliance checks.

Technical Skills

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- **Languages:** Java, C++, Python, Go, JavaScript, SQL
 - **Frameworks & Libraries:** Spring Boot, JavaFX, Selenium, Unity
 - **Tools & Infrastructure:** AWS (EC2), Redis, GraphQL, REST, Git, Docker, Linux

Languages

Turkish (native) · English (near-native) · French (fluent, C1) · Dutch (beginner, A2)